Open Source Architecture Appendix

Chen Xingqiang · github.com/chenxingqiang · zone-based end-to-end system architecture

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

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GPU profiling & replay

YiRage
RuntimeFusion compiler

JAX-sklearn
transparent JAX acceleration

Secret-Learn
privacy-preserving ML

YouTube-DNN
two-tower recall

Research CLI
agent workflow

Questions-Gen
three-stage training

阅读约定

自上而下读取主数据流；虚线框表示架构分区；盒内仅列模块与职责。

蓝色实线表示主链路，紫色虚线表示回退或反馈，不以源码文件充当架构层。

TraceSmith — 系统架构 · GPU Trace 端到端数据流

Application → bindings → capture / state / format → replay & export → GPU backends

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APPLICATION · 用户入口

C++ / Python Application
嵌入业务进程

CLI
profile · record · view · export

External UI
Perfetto · Chrome

BINDINGS · Python / C++ 边界

_tracesmith.so
pybind11 native extension

CORE LIBRARY · C++17

Capture Module
CUPTIProfiler · MCPTIProfiler
Metal · Ascend · ROCm · eBPF

Common + Format
RingBuffer SPSC · StackCapture
SBTWriter / SBTReader

State Module
GPUStateMachine
TimelineBuilder · PerfettoExporter

Replay Module
ReplayEngine · StreamScheduler
DeterminismChecker · FrameCapture

CROSS-CUTTING · 集群与调试

replay / debug feedback

Cluster Module
GPUTopology · TimeSync
NCCLTracker

GDB / Tracy
RSPHandler · GPUDebugEngine
TracyExporter

GPU BACKENDS · 系统接口

NVIDIA
CUDA · CUPTI · nsys

MetaX
MACA · MCPTI · mcTracer

Apple / AMD
Metal · xctrace · ROCm

Huawei / Linux
CANN · msprof · eBPF

OUTPUTS · Trace 产物

trace.sbt
compact binary

trace.json / .perfetto
Perfetto timeline

graph.dot / ASCII
analysis views

主链：GPU event → lock-free ring → SBT/state → replay 或 Perfetto；Cluster / GDB 为旁路能力。

YiRage (YPK) — 系统架构 · RuntimeFusion 数据流

Inference engine owns scheduling/KV; RF owns fusion identity; search and execute use the same backend

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INFERENCE ENGINE · 调度与 KV 所有者

vLLM
continuous batch · PagedAttention

SGLang
RadixAttention · extend meta

Decode Metadata
block table · seq lens · SM budget

RUNTIMEFUSION · 每个 decode step

RF.step(inputs, meta)
select 0..N capsules

FusionCapsule
MLP gate_up + down
segment pipeline

Native Fallback
skip → engine native ops

SEARCH & PLAN · offline / warm-up

superoptimize(backend)
KN / TB graph search

FusionPlan / μGraph
profile · fingerprint verify

Persistent Cache
shape bucket restore

Ray Coordinator
optional distributed search

fallback / re-search

BACKEND MANAGER · 编译与执行边界

BackendRegistry
search + execute same backend

KNGraph / TBGraph
transpiler → kernels

yirage.core
Cython runtime

BACKENDS → HARDWARE

CUDA / MACA
NVIDIA GPU · MetaX C500/C550

ROCm / MPS
AMD GPU · Apple Silicon

Ascend / NKI
Huawei 910 · AWS Neuron

CPU / MLIR / Triton
x86 · ARM · extra targets

主链：Engine metadata → RF.step → FusionCapsule → BackendRegistry → target hardware。

JAX-sklearn (xlearn) — 五层架构 · 透明加速数据流

100% sklearn API → runtime injection → proxy dispatch → array conversion → JAX hardware

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USER CODE · sklearn-compatible API

import xlearn as sklearn
LinearRegression · KMeans · PCA

Estimator Calls
fit · predict · transform

COMPATIBILITY · xlearn/__init__.py

__getattr__
lazy module loading

_auto_jax_accelerate_module
runtime class injection

AcceleratorRegistry
algorithm implementation map

JAX ACCELERATION · xlearn/_jax

EstimatorProxy
fit-time dispatch

Universal JAX
JIT · vmap · accelerated estimator

sklearn Fallback
original estimator

graceful fallback

DATA MANAGEMENT · boundary conversion

NumPy → JAX
dtype · shape normalization

Policy / Thresholds
dataset size · platform · config

JAX → NumPy
sklearn-compatible outputs

HARDWARE ABSTRACTION · JAX backend

CPU
jaxlib

CUDA GPU
JAX XLA

TPU
JAX XLA

Apple Metal
experimental

代理根据算法、数据规模和配置选择 JAX；不满足条件或执行失败时回退原生 sklearn。

Secret-Learn — 六层架构 · 隐私计算数据流

sklearn-compatible API → privacy mode → JAX / SecretFlow compute → secure devices

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APPLICATION · 多方隐私数据

Healthcare / Finance
cross-party modeling

IoT / Research
distributed learning

SKLEARN API · 统一算法入口

secretlearn.*
fit · predict · transform

algorithm_classifier.py
classify fit signature / epochs

PRIVACY MODES · execution placement

FEDERATED LEARNING
PYU local compute
HEU secure aggregation

SECRET SHARING
SPU MPC · ABY3 / CHEETAH

SPLIT LEARNING
split model · encrypted activations

ACCELERATION & COMPUTE

JAX Acceleration
JIT · vmap · xlearn

SecretFlow Runtime
party orchestration

SECURE DEVICES

PYU
party-local plaintext

HEU
homomorphic aggregation

SPU
secure MPC

TEE
optional enclave

INFRASTRUCTURE & HARDWARE

SecretFlow
distributed secure runtime

JAX / XLA
CPU · GPU · TPU

sklearn
API compatibility

同一 sklearn 调用由 FL / SS / SL 决定数据驻留与执行设备；JAX 加速可公开的本地计算。

YouTube-DNN-RecSys — 系统架构 · 召回数据流

Python / Spark data preparation → TFRecords → dnn.py → user & item vectors → prediction

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DATA PROCESSING · 离线样本构建

Python Pipeline
data2tfrecords.py
read_tfrecords.py

Spark Pipeline
MakeDataOne.scala
MakeDataTwo.scala

FeatureBuilder
history + continuous features

STORAGE · 训练与模型产物

TFRecords
train / evaluation

Checkpoints
ckpt / eval / modelpath

MODEL TRAINING · src/python/models/dnn.py

Input
visit_items_index × 5
continuous_features × 16

Item Embedding
500 × 16 · average pooling

User Tower
Dense 64 → 32 → vector 16

Softmax Output
500 classes · Top-K

evaluation feedback

PREDICTION · serving paths

Python SavedModel
load_dnn_model.py

User Vector
PredictUserVector.scala

Item Table
ItemEmbeddingPredictor.scala

RECOMMENDATION OUTPUT

Candidate Recall
user vector × item embedding

Top-K Items
next-item recommendation

主链：行为历史 → TFRecords → embedding mean-pool → user vector → item table matching → Top-K。

Research CLI — 系统架构 · Agent 请求数据流

Terminal UI → core orchestration & routing → tools / MCP / research modules → model providers

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CLI FRONTEND · packages/cli

Interactive TUI
research.tsx · sessions

Commands
mcp · skills · hooks

ACP
acpSession.ts

CORE ORCHESTRATION · packages/core

Context & Memory
config.ts · memory.ts

modelRouterService
provider selection

Routing Strategy
default · classifier · approval

TOOLS & INTEGRATIONS

Workspace Tools
filesystem · shell

MCP
external service adapters

Research Modules
writing · bibliography · analysis

SDK / A2A / VS Code
embedded clients

tool result loop

MODEL PROVIDERS · provider config

Gemini / Vertex
hosted models

OpenAI / Anthropic
hosted models

Baidu / DeepSeek / Qwen
hosted models

RESULT STREAM

Tool Results
structured observations

Assistant Response
streamed to terminal

主链：prompt → packages/cli → packages/core router → provider / tools → structured result → terminal。

Questions-Gen — 系统架构 · 三阶段训练数据流

Training data → SFT → GRPO → DeepSeek distillation → validation → Hugging Face / Ollama

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TRAINING DATA · problem mixture

Historical Problems
50% source distribution

Variants
30% conditional variation

Novel Problems
20% innovation target

Stage 1 SFT · QuestionsGenTrainer.py

Qwen3-14B
supervised fine-tuning

Stage-1 Checkpoint
stage1-fp-merged

Stage 2 GRPO · reward optimization

RewardCalculator
0.4 difficulty · 0.3 novelty
0.2 rigor · 0.1 diversity

NoveltyConstraint
variation / duplication guard

GRPO Trainer
group-relative optimization

Stage 3 Distill · teacher supervision

quality feedback

DeepSeekTeacher
DeepSeek-R1 reasoning teacher

Final Model
final-merged-16bit

VALIDATION

ModelValidator
single checkpoint

BatchValidator
algebra · geometry · calculus

QualityEvaluator
clarity · difficulty · originality

DEPLOYMENT

Hugging Face
three published stages

Ollama
local inference

questions-gen CLI
validate · batch · quality

主链：混合题库 → SFT → GRPO reward/novelty → DeepSeek-R1 distill → validate → publish。